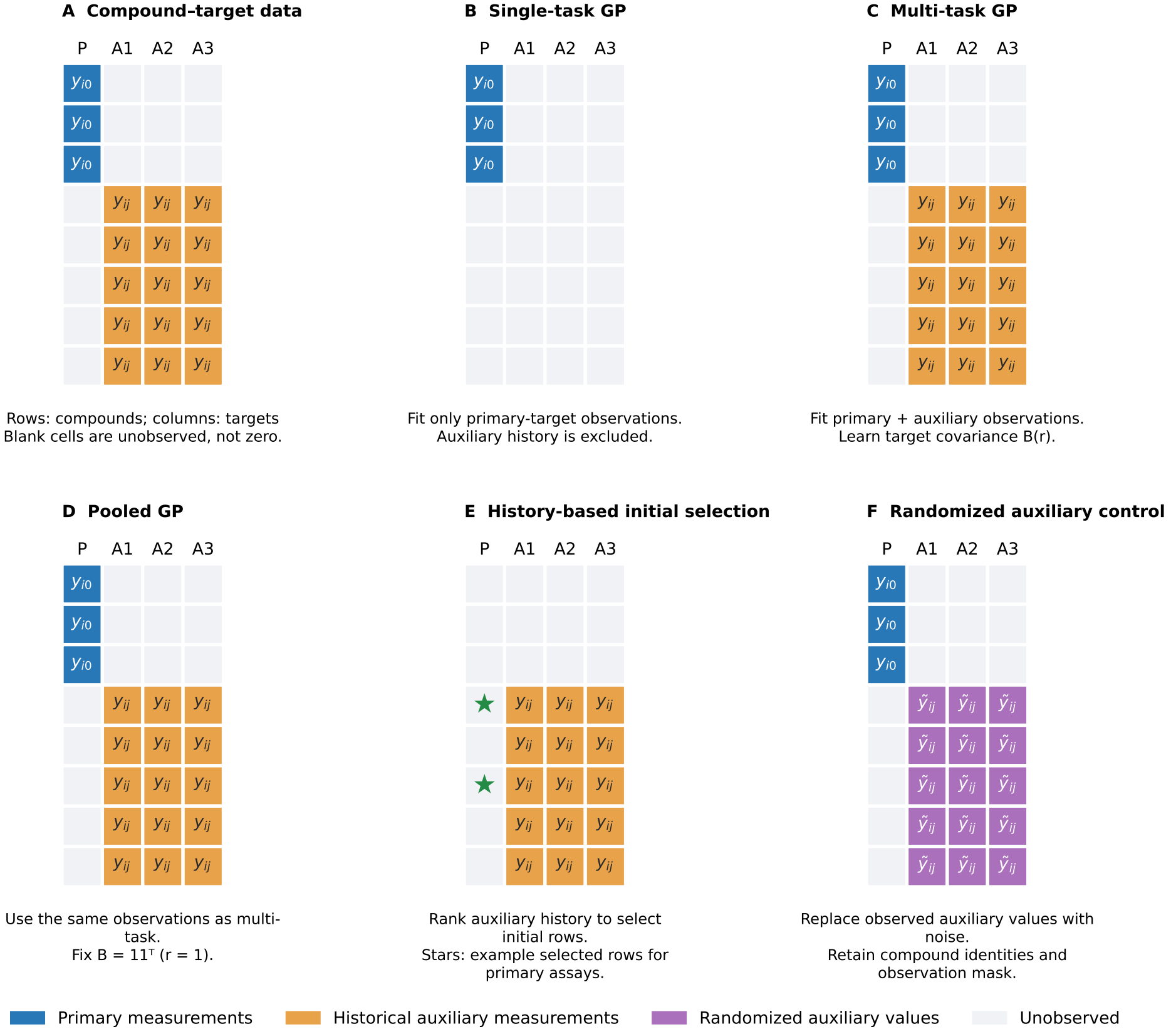
A Compound–target data
P A1 A2 A3
Rows: compounds; columns: targets
Blank cells are unobserved, not zero.
B Single-task GP
P A1 A2 A3
Fit only primary-target observations.
Auxiliary history is excluded.
C Multi-task GP
P A1 A2 A3
Fit primary + auxiliary observations.
Learn target covariance B(r).
D Pooled GP
P A1 A2 A3
Use the same observations as multi-task.
Fix B = 11ᵀ (r = 1).
E History-based initial selection
P A1 A2 A3
Rank auxiliary history to select initial rows.
Stars: example selected rows for primary assays.
F Randomized auxiliary control
P A1 A2 A3
Replace observed auxiliary values with noise.
Retain compound identities and observation mask.
Primary measurements
Historical auxiliary measurements
Randomized auxiliary values
Unobserved